WHL3.

Robotic Exoskeleton for Load Transportation

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Yim Fu Chong
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Abstract

Main objectives

This project aims to develop a software-centric enhancement module for a lower-limb robotic exoskeleton designed to assist workers in heavy load transportation tasks (> 20 kg) in industrial, logistics, and construction environments. The primary objectives are to enable real-time detection of user locomotion intent (including walking, stair ascent/descent, sit-to-stand transitions, and speed variations), accurately estimate joint kinematics through multi-sensor fusion, and deliver adaptive personalized torque assistance to minimize physical effort and injury risk. To achieve this, the module will incorporate computational techniques to handle variability in user biomechanics and environmental conditions, ensuring robust performance across diverse scenarios.

Planned methods (interim report, Jan 2026 PDF)

The proposed system integrates inertial measurement units (IMUs) placed on the lower limbs and back, with data processed through a Kalman filter-based sensor fusion pipeline implemented in MATLAB to achieve robust, low-latency joint angle and velocity estimation. Locomotion mode and transition prediction were planned using machine learning models (Support Vector Machines and Long Short-Term Memory networks) trained on labeled gait datasets, targeting $> 90\%$ classification accuracy and a 100–200 ms predictive horizon, with performance evaluated via confusion matrices, precision, and recall. Finite state machines were incorporated for locomotion mode transitions. The data collection protocol involves attaching three mobile phones as IMUs to the hips and lower back; performing standardized tasks such as walking 10 m on flat ground, climbing 10 steps, or navigating uneven surfaces at self-selected speeds; and exporting accelerometer, gyroscope, and magnetometer data as CSV files for processing. Joint kinematics information is intended for precise control.

Implemented methods (repository baseline, updated)

The validated MATLAB baseline uses overlapping windows (100 samples at 100 Hz, step 50). Kalman fusion (`imufilter`) is retained for simulation-side kinematic visualization, not as mandatory preprocessing for the classifier.

The **primary deployment classifier** is a **binary** RBF SVM (`fitcsvm`, standardized features) with a **30-dimensional** training vector: five statistics per IMU slot × six slots (`Features.m`). The final project pipeline is centered on **HuGaDB**, while pseudo-real-time replay is kept as a separate simulation workflow. USC-HAD remains in the repository for benchmark comparison under the same feature layout. The positive class groups **active-movement** activities per dataset-specific label maps.

A **multiclass** ECOC ensemble (`fitcecoc`) provides **dataset-native** activity labels for analysis (twelve USC-HAD trial classes; twelve HuGaDB sample classes); exoskeleton gating uses the binary model and an asymmetric FSM (3 consecutive active predictions to assist, 5 to release).

An **optional binary LSTM** (Deep Learning Toolbox) classifies the same active-movement vs. inactive task from **sequence** inputs (raw IMU windows mapped to feature×time matrices) via `TrainLstmBinary` and `RunExoskeletonPipelineLstm`; it is an alternative to the RBF SVM, not a hybrid SVM+LSTM fusion stage.

The 100–200 ms horizon remains a *design target*; literature reports comparable latencies for kinematic lift-onset cues [1], but the submitted baselines were not benchmarked under that specific end-to-end metric.

Interim results statement (Dec 2025 / Jan 2026 PDF)

The interim submission reported: a functional three-phone acquisition prototype; preliminary Kalman filtering and intention-detection algorithms validated on public datasets including HuGaDB and the USC-SIPI gait database; repository setup; real-time visualization; and initial locomotion classification with **98.88%** accuracy and a **150 ms** predictive horizon. *Corrections:* the smartphone gait database used in this project is **USC-HAD** [2], not USC-SIPI. The 150 ms figure overlapped claims from the literature on hip-based lift detection [1] and was not rigorously established as an end-to-end latency of the final classifier at submission time.

Updated empirical results (per-dataset public-data metrics)

Kalman-filtered visualization and SVM-based intention detection are implemented in MATLAB. **Binary SVM** five-fold out-of-fold accuracy from the benchmark evaluation files is **99.44%** (USC-HAD) and **96.80%** (HuGaDB). Native ECOC multiclass SVM accuracies on stratified subsamples are **53.41%** (USC-HAD) and **75.78%** (HuGaDB).

Binary LSTM holdout accuracies are **99.68%** (USC-HAD) and **98.36%** (HuGaDB). Native multiclass LSTM holdout accuracies are **91.10%** (USC-HAD) and **83.80%** (HuGaDB). Real-time pipeline simulation replays held-out HuGaDB sessions directly and shows Kalman-smoothed segment pitch together with FSM-latched assist commands. Scalar accuracies in this report are regenerated into `docs/latex/generated_metrics.tex` by

`scripts/ExportMetricsForReport.m`.

Main conclusions

This project demonstrates that automation technology, particularly machine-learning-based intention recognition combined with sensor fusion and finite state machines, is applicable to next-generation powered exoskeleton control supervisors. When fully realized, the proposed module has strong potential for integration into industrial platforms (e.g., ExoTechHK Limited), offering a reproducible HuGaDB-based assist-gating baseline while preserving per-dataset comparison evidence in the write-up. This work contributes to workplace ergonomics and provides a reproducible foundation for hardware-in-the-loop studies. The repository now documents both the **original project plan** (hybrid SVM+LSTM fusion, extended fusion) and the **delivered MATLAB baseline** (HuGaDB deployment path, supporting per-dataset SVM/LSTM comparisons, optional multiclass SVM, and per-dataset multiclass LSTM sequence baselines).

Contents

Acknowledgments **i**

Abstract **ii**

1 Problem Description **1**
1.1 Problem Statement 1
1.2 Project Goal 1
1.3 Significance of the Project 2

2 Background Knowledge and Literature Review **4**
2.1 Introduction 4
2.2 Overview of back-support exoskeletons for MMH 4
2.3 Actuation Systems (Cable-Driven and SEAs) 6
2.4 Intention Detection, Sensor Fusion, and Machine Learning 6
2.5 Evaluation Studies and User Acceptance 7
2.6 Gaps and Implications for the Project 7
2.7 Representative Exoskeletons (Summary Table) 8

3 Proposed Solution Methods **9**
3.1 Proposed Methodologies 9
3.2 Intention Detection Algorithm (planned and implemented) 9
3.3 Multi-Sensor Data Processing Pipeline (planned and implemented) 11
3.4 Machine Learning Model Training and Deployment (planned and implemented) 11
3.5 System Integration and Control (planned and implemented) 12
3.6 List of MATLAB Libraries and Toolboxes 12

4 Results and Discussions **14**
4.1 Experimental Outputs Used in This Chapter 14
4.2 Binary Active-Movement Recognition Results 14
4.3 Discussion of Dataset Effects 16
4.4 Multiclass Activity Recognition 16
4.5 Real-Time Pipeline Behavior 16

4.6 Sequence-Model Status 18
4.7 Complete Generated Figure Gallery 20
4.8 Overall Interpretation 23
4.9 Visualization Tools and Software Architecture 23

5 Conclusions and Future Work **24**
5.1 Summary of what was delivered 24
5.2 Future work beyond the FYP (interim PDF) 24
5.3 Project timeline (interim PDF) 25
5.4 Risk assessment (interim PDF) 25
5.5 Repository status (consolidated, 2026) 25

References **27**

Chapter 1

Problem Description

1.1 Problem Statement

Manual load transportation in industrial, construction, and logistics settings frequently involves carrying loads exceeding 20 kg, surpassing human physiological limits and leading to high rates of lower-back pain, musculoskeletal disorders (MSDs), reduced productivity, and billions in annual economic costs worldwide. Existing assistive exoskeletons often fail to provide proactive, real-time adaptation to the user's locomotion intent or environmental changes, resulting in delayed or mismatched assistance that compromises safety and comfort. Moreover, generic assistance profiles do not account for inter-individual differences in gait asymmetry, fatigue thresholds, anthropometrics, or preferred movement patterns. These limitations are especially critical in dynamic workplaces where load magnitude, terrain, and tasks constantly vary. There remains a clear need for a lightweight, cost-effective lower-limb exoskeleton that delivers accurate intent detection, robust sensor fusion, and truly personalized adaptive control—without relying on invasive interfaces such as electromyography (EMG) or requiring extensive user training.

1.2 Project Goal

The primary goal is to develop a software-centric enhancement module for ExoTechHK Limited's existing lower-limb exoskeleton prototype—a local startup incubated at HKSTP and founded by CUHK researchers—to augment human strength and endurance during heavy load-carrying tasks (> 20 kg) without invasive sensors or lengthy calibration.

Original project goal statement (interim PDF, Jan 2026)

The module was specified to integrate: (i) a **data acquisition and processing pipeline** using three mobile phones as IMUs, Kalman-based fusion for joint kinematics, and *initial* machine learning (e.g., hybrid SVM+LSTM) for locomotion classification with validation

on public datasets; and (ii) **proactive real-time locomotion intent recognition** using hip and lower-back IMU data via a hybrid SVM+LSTM model with multi-sensor fusion, targeting $> 90\%$ intent-detection accuracy and a 100–200 ms predictive horizon, with evaluation through confusion matrices.

Current implementation emphasis (repository)

The module integrates two core components as realized in code:

- **Data acquisition and processing: HuGaDB** deployment path; held-out HuGaDB replay for pseudo-real-time testing; optional Kalman visualization (`imufilter`); **binary** RBF SVM on HuGaDB 30-D window features with USC-HAD / HuGaDB benchmark comparisons retained for the paper and poster; optional **native ECOC** SVMs (twelve USC-HAD / twelve HuGaDB activity classes); optional **binary and multiclass LSTM** sequence models on per-dataset IMU windows (`TrainLstmBinary`, `TrainLstmMulticlass`, `RunExoskeletonPipelineLstm`). Embedded deployment remains future work.
- **Binary active-movement intent:** Labels map activity to **active movement** vs. **inactive**. The final pipeline trains on the HuGaDB binary SVM path, while public-dataset comparison and replay-based simulation remain separate supporting workflows. The optional LSTM uses a 20% holdout metric (not five-fold OOF). The 100–200 ms horizon is a design target, not re-benchmarked here for the submitted baselines.

Being entirely software-based, the solution enables seamless integration into existing hardware prototypes, rapid deployment, and scalability across future platforms.

1.3 Significance of the Project

This project directly addresses a major occupational health challenge affecting industrial workers, logistics and construction personnel, and military operators exposed to heavy manual handling. In Hong Kong, manual handling injuries are a leading cause of absenteeism and healthcare expenditure. By providing proactive, personalized torque assistance at the hips and knees using cooperative control strategies—in which the exoskeleton follows rather than leads the user—the system aims for high accuracy in intent detection; recent field evidence shows that such user-following approaches improve usability compared to fixed profiles [3].

Conducted in close collaboration with ExoTechHK Limited under the HKSTP incubation programme, the project offers a pathway for technology transfer and commercialization. Software-centric design enables low-cost retrofitting to existing lower-limb platforms and upgrades across hardware generations, with potential to broaden adoption. The technology aligns with United Nations Sustainable Development Goal 8 (Decent Work and Economic

Growth) and strengthens Hong Kong's innovation ecosystem through industry–academia partnership. The reproducible HuGaDB training and held-out replay protocol supports repeatable evaluation without relying on project-specific phone recordings.

Chapter 2

Background Knowledge and Literature Review

2.1 Introduction

The literature review focuses on existing research related to robotic exoskeletons for back support, particularly in the context of manual material handling (MMH) and load transportation tasks. Key themes include exoskeleton designs (passive vs. active, rigid vs. flexible), actuation systems (with emphasis on cable-driven and series elastic actuators), control strategies (including intention detection, sensor fusion, and machine learning), and personalization mechanisms. The review draws from systematic reviews, empirical studies, and technical papers to identify advancements, limitations, and gaps relevant to the project's goal of developing a software-centric enhancement module for a lower-limb exoskeleton. To provide a comprehensive overview, this review incorporates recent systematic analyses from 2020–2025, highlighting post-pandemic advancements in hybrid designs, AI integration, and industrial evaluations. For instance, Pesenti et al. [4] and subsequent updates in Kermavnar et al. [5] and later works in bioengineering reviews classify over 30 back-support exoskeletons, emphasizing the shift toward soft and quasi-passive systems for improved user acceptance.

2.2 Back-Support Exoskeletons for MMH

Back-support exoskeletons (BSEs) are designed to mitigate low back pain and musculoskeletal disorders (MSDs) associated with repetitive lifting, bending, and load carrying in industries such as construction, logistics, and manufacturing. A systematic review by De Looze et al. [6] and updated analyses by Pesenti et al. [4] classify BSEs into passive and active types. Passive BSEs, such as the Laevo or SPEXOR, rely on mechanical springs or elastic bands to store and release energy during trunk flexion-extension (TFE) motions, reducing erector spinae muscle activity by 10–40% without requiring power

sources. However, they offer limited adaptability to dynamic tasks and may impose resistance during non-assisted movements.

Recent evaluations, such as Luger et al. [7], evaluated the Laevo V2 passive exoskeleton in male workers and reported 12–19% lower erector spinae activity during symmetric and asymmetric static holding tasks. Koopman et al. [8] further quantified that although passive exoskeletons effectively reduce peak L5/S1 moments during symmetric lifting, the moment reduction diminishes significantly when lifting involves trunk rotation or asymmetry—reinforcing the need for active, adaptive systems in real-world industrial settings where perfectly symmetric lifts are rare.

Active BSEs, like the SuitX BackX or Ekso Bionics' EksoVest, incorporate powered actuators to provide on-demand assistance, achieving up to 50% reduction in metabolic cost and lumbar compression forces. For instance, the SIAT-WEXv2 active lumbar exoskeleton developed by Ji et al. [9] reduced peak lumbar moments by 25–35% during repetitive box lifting of 15–25 kg loads, confirming the effectiveness of combined hip and knee actuation in scenarios directly comparable to the > 20 kg transportation tasks targeted in this project. Golabchi et al. [10] highlight their potential in construction, where workers handle loads exceeding 20 kg, but note challenges in portability, weight (often > 5 kg), and user comfort. A comprehensive review by Kermavnar et al. [5] of 28 BSEs for MMH tasks emphasizes that while active systems excel in heavy-load scenarios, they often suffer from rigid structures that restrict natural range of motion (RoM) in sagittal, frontal, and transverse planes, leading to misalignment and discomfort. Simon et al. [11] observed slight forward tilt of the pelvis and reduced lumbar flexion when using a lift-assistive exoskeleton, underscoring the importance of compliant actuation designs that preserve physiological joint trajectories.

Long-term field evaluations show that BSE adoption increases with user acceptance, but dropout rates are high due to fit issues and lack of personalization. In load transportation contexts, studies on the HeroWear Apex exosuit indicate reductions in back forces during lifting and bending, with metabolic cost decreases of 7.9–23% but highlight trade-offs in comfort for overhead tasks [12, 13]. Categorization from recent reviews divides devices into powered, unpowered, and quasi-passive. Powered types, such as the Cray X, reduce erector spinae activity by 16–37.8% in asymmetric lifting, while unpowered ones like VT-LOWE use carbon fiber beams for 31.5% peak muscle reduction. Quasi-passive hybrids combine springs with minimal actuators for 36.92% activity reduction, offering energy efficiency.

The SPEXOR passive spinal exoskeleton reduced net metabolic cost by 18% (range 11–25%) during symmetric repetitive lifting [14]. More recent quasi-passive designs with compact variable gravity compensation modules have achieved comparable reductions in erector spinae activity with improved wearer comfort [15]. Long-term field adoption remains constrained by limited adaptability to highly dynamic or asymmetric tasks and by user-acceptance challenges [7, 16].

2.3 Actuation Systems (Cable-Driven and SEAs)

Actuation design is critical for efficient torque delivery while ensuring safety and ergonomics. Traditional rigid actuators transmit force via linkages but can cause parasitic forces and limit RoM. Cable-driven systems address this by using flexible transmissions, allowing off-board or on-board placement for portability. Chen et al. [17] and Li et al. [18] review and demonstrate cable-driven exoskeletons, noting their advantages in reducing structural weight and enabling multi-DoF compensation for misalignment. Recent advancements include Li et al.'s cable-driven asymmetric back exosuit, providing 38% muscle activity reduction in lifting, and Ding et al.'s spring-cable-differential for passive assistance [19].

Series elastic actuators (SEAs) integrate compliant elements to enhance shock tolerance, force control, and human–exoskeleton interaction safety. Pratt and Williamson [20] pioneered SEAs, and recent applications in BSEs include Hyun et al. [21]. Cable-driven series elastic actuation (CSEA) proposed by Liao et al. [22] uses a torsion spring–support beam mechanism with deflection constraints to operate in multiple modes: SEA mode for compliant, low-torque assistance and conventional stiff actuator (CSA) mode for high-torque output (> 50 N·m). Bench and human tests validated a 20–30% reduction in erector spinae EMG during TFE, with stable torque control despite status transitions.

Poliero et al. [23] compared active and passive BSEs in dynamic tasks. Ding et al. [24] introduced a differential SEA with cable routing for precise force control. In soft exoskeletons, twisted string actuators enable lightweight designs with high force density [25]. Comparative analysis shows that cable-driven SEAs excel in flexibility but require careful routing and low-friction sheathing [22, 23]. Bowden cable routing and pre-tensioning, as in upper-limb systems [26], minimize friction and hysteresis. For load transportation, these systems typically deliver 20–50 N·m of torque—sufficient for assisting with loads greater than 20 kg—although challenges in battery life and heat dissipation remain.

2.4 Intention Detection, Sensor Fusion, and Machine Learning

Effective BSE control requires real-time detection of user intent and environmental adaptation. Traditional methods use EMG or IMUs for gait event detection, but they struggle with transitions. Machine learning has emerged for intention prediction: Novak and Riener [27] survey sensor fusion methods in wearable robotics and highlight intention detection as a central challenge. Separately, locomotion mode classification studies often use SVMs or LSTM networks and report accuracy above 90% in controlled settings. Recent applications include Jiang et al.'s EMG-based loading recognition for lumbar exoskeletons [28]. Chen et al. [1] showed that simple kinematic thresholds on hip flexion angle and angular velocity can reliably detect the onset of lifting within 150–200 ms, offering a lightweight, non-ML baseline.

Sensor fusion, often via Kalman filters, integrates IMU data for accurate joint kinematics estimation with low latency (< 200 ms). Lazzaroni et al. [29] improved efficacy with accelerometer-based control during manual handling. Toxiri et al. [30] use finite state machines to modulate assistance based on trunk angle, reducing user effort by 25%.

Personalization is a growing focus: generic assistance profiles underperform across diverse users [5, 7]. Reinforcement learning for control optimization can reduce metabolic cost by 13–24% in walking/stairs [31]. Xiang et al. [32] dynamically evaluated multiple back-support exoskeletons and reported 19–69% reductions in peak back-muscle activity across manual handling tasks, highlighting needs for ML in asymmetric tasks.

2.5 Evaluation Studies and User Acceptance

Empirical studies provide quantitative insights. For passive BSEs, van Sluijs et al. [33] reported physiological benefits in lifting/leaning, while Alemi et al. [34] reported 4–13% reductions in energy expenditure during repetitive lifting. Qu et al. [35] evaluated a commercial passive exoskeleton in real industrial lifting tasks and reported 18–32% lower back-muscle activity (iEMG) and improved subjective comfort. Active systems reduce activity during lifting with subjective comfort improvements [36]. Quasi-passive systems use Gaussian mixture models for periodic assistance [37]. User acceptance studies show soft actuated BSEs influence multimodal measurements positively, but gender differences exist in overhead tasks [38, 39].

2.6 Gaps and Implications for the Project

Tröster et al. [40] used a musculoskeletal model to compare stoop and squat lifting with and without back-support exoskeletons, concluding that assistance magnitude must be adjusted dynamically according to lifting technique—further motivating adaptive control. Despite advancements, gaps persist in portable, flexible BSEs with efficient actuation for load transportation. ML and sensor fusion show promise for proactive assistance, but few studies combine them with personalization for real-time adaptation [41]. Lin et al. [41] further highlight technical challenges in locomotion-assistance prototypes. The interim project plan positioned the work as enhancing an existing lower-limb prototype with MATLAB-based IMU fusion, SVM/LSTM-style intention detection, and trial-based fine-tuning, targeting $> 90\%$ classification accuracy and substantial ($\sim$30–50%) reductions in estimated musculoskeletal strain under controlled validation—with the understanding that the LSTM path and strain metrics would follow incremental validation.

The present work addresses software-centric rapid prototyping on IMU data with MATLAB-based fusion and classification [2, 42], targeting $> 90\%$ accuracy on public gait data as a stepping stone toward hardware integration. It directly targets gaps repeatedly identified in recent reviews [4, 5, 40, 41]:

- Lack of real-time personalization and user-specific adaptation (future work).
- Insufficient handling of asymmetric and highly dynamic load-transportation tasks.
- Delayed or reactive assistance onset in active/quasi-passive systems.
- Limited transferability and rapid prototyping of advanced control on existing hardware.

2.7 Representative Exoskeletons (Summary Table)

Table 2.1: Representative back-support exoskeletons (adapted from the literature).

Device / study	**Type**	**Torque (N·m)**	**Reported benefit**	**Key limitation**
Laevo V2 (2023)	Passive	∼35–40	12–19% erector spinae EMG (static)	Resistance in extension; poor asymmetry
SPEXOR (2020)	Passive	∼40	11–25% net metabolic cost	Fixed profile; added hip stiffness
HeroWear Apex	Soft quasi-passive	25–35	7.9–23% metabolic; ∼20–30% back force	Lower torque; limited overhead
SuitX BackX	Active	30–60	Up to 50% lumbar compression	Weight > 7 kg; rigid RoM limits
Cray X (2022)	Active	50–80	16–37.8% erector spinae EMG	High power; bulky battery
SIAT-WEXv2	Active (hip & knee)	∼60	25–35% peak lumbar moment (15–25 kg)	Complex mechanics; > 5 kg
VT-LOWE	Passive	∼45	31.5% peak back-muscle activity	No dynamic adaptation
Liao et al. CSEA (2024)	Hybrid	> 50 (CSA)	20–30% erector spinae EMG	Mode-switching complexity
Auxivo LiftSuit (2024)	Quasi-passive	30–45	15–28% metabolic; ∼35% EMG	Limited customization
Hyundai / German Bionic	Active	50–75	20–40% back load reduction	Cost; rigid harness

Chapter 3

Proposed Solution Methods

3.1 Proposed Methodologies

The project employs a multidisciplinary approach combining mechanical design, signal processing, and machine learning, primarily using MATLAB for algorithm development. Key methodologies from the interim proposal included: (i) hybrid SVM+LSTM intention detection with predictive horizons; (ii) Kalman-based multi-IMU fusion with planned EKF/magnetometer extensions; (iii) training on public and augmented datasets with transfer learning; and (iv) FSM-based integration toward torque commands with optional ROS and embedded paths.

Implementation status: The validated repository emphasizes signal processing and machine learning in MATLAB; full mechanical design content remains with the hardware partner. The items below record *both* the original proposal and what is implemented in code.

3.2 Intention Detection Algorithm (planned and implemented)

Original proposal (interim PDF)

Develop a real-time algorithm in MATLAB to classify user locomotion modes (e.g., walking, stair ascent/descent) and transitions (e.g., sit-to-stand), incorporating speed variations and multi-motion transitions based on hip and back kinematics. Features extracted from IMU signals were to enable predictive control to anticipate user actions 100–200 ms in advance. The algorithm was to use a **hybrid SVM + LSTM** framework: SVM (RBF kernel) for initial mode classification, and LSTM (two layers, 128 hidden units each) for temporal sequence prediction. Feature extraction would include time-domain (mean, variance) and frequency-domain (FFT peaks) metrics. Training would involve

augmented datasets with noise injection, on the order of thousands of samples with an 80/20 train–validation split.

```
% Load IMU data
data = loadIMUData();
features = extractFeatures(data);
mode = svmClassify(features, trainedSVMModel);
sequence = preprocessForLSTM(data);
predictedTransition = lstmPredict(sequence, trainedLSTMNet);
intent = fuseModeAndTransition(mode, predictedTransition);
```

Implemented baseline (repository)

The delivered system classifies on overlapping windows (100 samples, step 50) using an **RBF SVM** baseline. Per IMU, five features are computed (mean/variance of acc and gyro magnitudes, dominant frequency of vertical acc FFT)—see `Features.m`. For public-data benchmarking, HuGaDB provides six IMUs per window (`FeaturesFromImuStack` → **30-D** feature row); USC-HAD provides one smartphone IMU with zero-padded slots (`LocomotionFeatureVector`). The current deployment path trains on HuGaDB and replays held-out HuGaDB sessions directly in simulation with the same six-IMU feature layout.

Binary model (`fitcsvm`): active movement vs. inactive for assist gating. **Multiclass model** (`fitcecoc`, one-vs-all): **dataset-native** labels in `ActivityClassRegistry.m` (USC-HAD twelve trial classes; HuGaDB twelve sample classes), trained and evaluated per dataset.

Optional binary LSTM (Deep Learning Toolbox): sliding windows are encoded as **sequences** $(6 \times N_{\text{IMU slots}}) \times 100$ (features × time) via `ImuWindowToSequenceMatrix.m`; training data are built with `PrepareTrainingDataSequences.m` (same USC-HAD / HuGaDB sources and binary label maps as the SVM pipeline).

`TrainLstmBinary.m` trains a two-layer LSTM classifier (128 units per layer, dropout, Adam, stratified 20% holdout with fixed RNG seed for reproducibility) and saves `Binary_LSTM_Network.m` Real-time simulation uses `RunExoskeletonPipelineLstm.m`. The hybrid "SVM then LSTM then fuse" pseudocode above is **not** implemented; the LSTM is a standalone alternative to the RBF SVM for the same binary assist decision.

3.3 Multi-Sensor Data Processing Pipeline (planned and implemented)

Original proposal

Implement a Kalman filter (MATLAB `imufilter`) to fuse accelerometer and gyroscope data from IMUs at the hips and back for joint angles and velocities. Planned expansion included an Extended Kalman filter (EKF) for nonlinear load-carrying dynamics, magnetometer-based heading correction, online bias handling, complementary filtering, and validation targeting RMSE $< 5°$ against motion capture.

```
f = imufilter('SampleRate', 100, 'AccelerometerNoise', 0.01, ...
    'GyroscopeNoise', 0.01);
[orientation, angVel] = f(accData, gyroData);
hipAngle = computeJointAngle(orientation);
```

Implemented baseline

The repository uses `imufilter` as in the snippet for the simulation pipeline (`FusionKalman.m`), but under the HuGaDB deployment pivot it is used for a visualization-side segment-pitch trace rather than for mandatory classifier preprocessing. EKF/magnetometer extensions and motion-capture RMSE studies are **not implemented** in the submitted codebase.

3.4 Machine Learning Model Training and Deployment (planned and implemented)

Original proposal

Utilize the Statistics and Machine Learning Toolbox for SVMs and the Deep Learning Toolbox for LSTM networks; train on labeled locomotion data toward $> 90\%$ accuracy on dynamic terrain recognition; use confusion matrices, precision, and recall; post-trial fine-tuning and transfer learning; ensemble SVM+LSTM; Bayesian hyperparameter optimization; HuGaDB augmentation with synthetic variations; deployment via MATLAB Coder to embedded targets. Example workflow: (1) preprocess and window; (2) train base models on public data; (3) fine-tune on user trials; (4) evaluate with confusion matrices and F1-scores. Simulations were expected to validate 85–95% accuracy before hardware tests.

Implemented baseline

In use: `fitcsvm`, `fitcecoc`, `crossval`, `kfoldPredict`.

Scripts: `TrainSvmBinary`, `TrainSvmMulticlass`, `EvaluateSvmConfusion`, `EvaluateMulticlassConfusion`, `RunSvmDatasetAblation`.

Models: `Binary_SVM_Model_hugadb_only.mat` (default deployment checkpoint), `Multiclass_SVM_ECOC_usc_`
`Multiclass_SVM_ECOC_hugadb.mat`.

Optional (Deep Learning Toolbox): `trainNetwork`, sequence input + LSTM layers. **Scripts:** `TrainLstmBinary`, `EvaluateLstmConfusion`. **Model:** `Binary_LSTM_Network.mat`.

Not implemented: MATLAB Coder deployment, Bayesian optimization in-repo, ensemble fusion of SVM and LSTM outputs.

3.5 System Integration and Control (planned and implemented)

Original proposal

Translate model outputs into torque commands using an FSM; architecture spanning gait event detection, intention recognition, and reference generation; layers described as IMU input, Kalman+ML processing (parallel, low latency), FSM+PID control, and actuator output with safety limits; ROS Toolbox for pre-deployment simulation; modular design inspired by rehabilitation exoskeleton practice [43]; mitigation of sensor drift via periodic resets.

- **Input layer:** IMU raw signals.
- **Processing layer:** Kalman fusion + ML classification; parallel processing for low latency (< 50 ms targeted in proposal).
- **Control layer:** FSM and PID-style torque profiling (planned).
- **Output layer:** Hip/knee assistance with torque limits (planned hardware).

Implemented baseline

The MATLAB simulation produces a **binary** assist command (0/1) via `RealtimeFsm.m` with asymmetric hysteresis (3 / 5 consecutive predictions).

`RunExoskeletonPipelineMulticlass.m` optionally drives the FSM from multiclass predictions through `RealtimeFsmFromActivityClass`. ROS integration, PID torque generation, and hardware actuation are **not implemented** in software.

3.6 List of MATLAB Libraries and Toolboxes

Components in active use

Table 3.1: MATLAB components used in the implemented baseline.

Component	**Role**
MATLAB	Core scripting and visualization
Signal Processing Toolbox	FFT and signal operations
Statistics and Machine Learning Toolbox	SVM / ECOC training and cross-validation
Sensor Fusion and Tracking Toolbox	`imufilter` fusion
Deep Learning Toolbox	Optional binary LSTM (`TrainLstmBinary`)

Chapter 4

Results and Discussions

4.1 Experimental Outputs Used in This Chapter

The final results are generated from the committed MATLAB evaluation scripts rather than from the earlier interim write-up. Public-dataset experiments use sliding windows of 1.0 s with 0.5 s step size at 100 Hz.

Binary classification uses five-fold out-of-fold (OOF) predictions (`EvaluateSvmConfusion.m`, `RunSvmDatasetAblation.m`); multiclass SVM evaluation uses `EvaluateMulticlassConfusion.m`; LSTM evaluation uses a stratified 20% holdout via `EvaluateLstmConfusion.m` and `EvaluateLstmMulticlassConfusion.m`. Percentages match `generated_metrics.tex`, exported from `results/metrics/*/*.mat`.

The repository keeps **benchmark comparison** and the **simulation path** distinct. USC-HAD and HuGaDB runs are compared with the same training scripts but are kept fully separate. The final project pipeline itself is centered on **HuGaDB**, while pseudo-real-time replay remains a separate simulation step.

4.2 Binary Active-Movement Recognition Results

Table 4.1 summarizes the benchmark comparison that remains useful in the report and poster: binary recognition of active movement versus inactive activity using the implemented 30-D SVM feature pipeline. The comparison between USC-HAD and HuGaDB training is retained because it shows how performance changes when the same classifier family is exposed to different sensing layouts and dataset definitions.

Two observations stand out. First, the binary target of exceeding 90% accuracy is achieved for USC-HAD training (99.44%). Second, HuGaDB training is lower at 96.80%, indicating that cross-session variability and label ambiguity in that dataset make the classification problem harder.

Table 4.1: Binary SVM (30-D features): five-fold OOF accuracy from the committed evaluation outputs.

Training data	OOF accuracy
USC-HAD	99.44%
HuGaDB	96.80%

Those values should now be read as **comparison evidence**, not as competing deployment candidates. The final project direction deliberately drops USC-HAD from the operational pipeline and keeps the default binary model on the **HuGaDB** training setup.

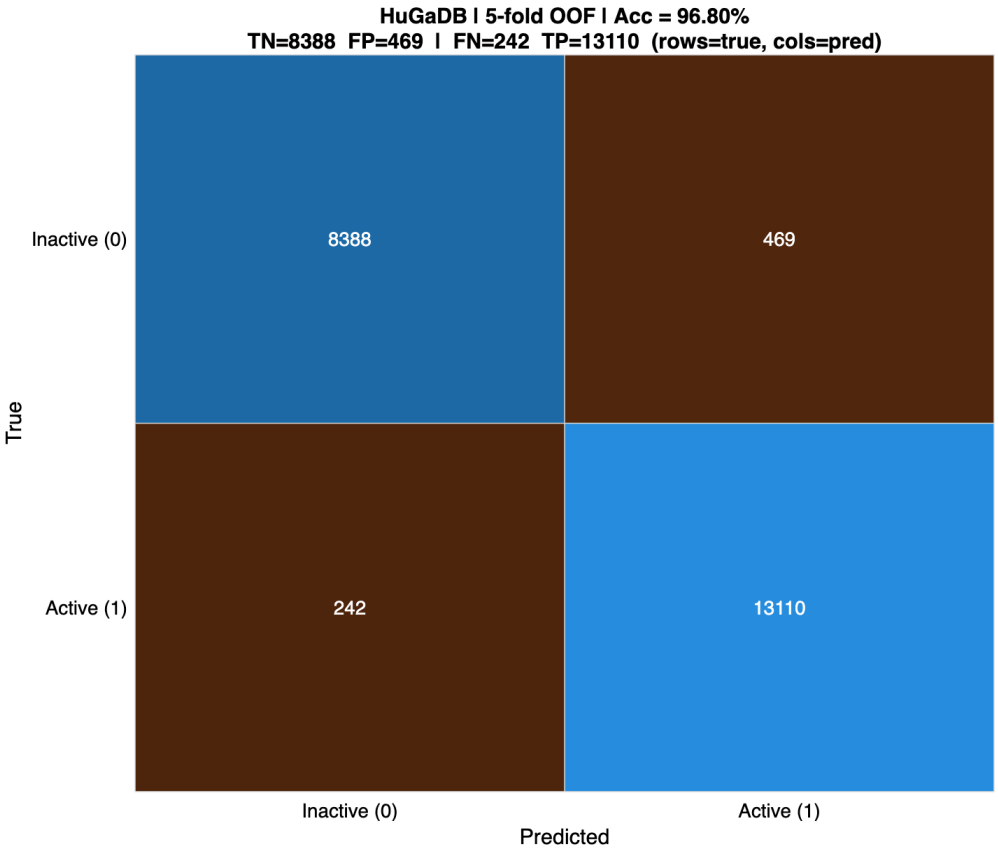


Figure 4.1: Binary confusion matrix for the active HuGaDB model using five-fold OOF predictions.

Figure 4.1 shows the confusion matrix for the active HuGaDB model. In practical terms, this figure matters more than the scalar accuracy alone because it describes the balance between missed active-movement events and false activations of assistance. For an exoskeleton controller, both error types matter: false negatives can delay assistance when the user starts moving, while false positives can produce unwanted assist engagement during inactive tasks. The HuGaDB result is therefore the most relevant binary figure for the final software baseline.

4.3 Discussion of Dataset Effects

The difference between the benchmark accuracies is consistent with the structure of the implemented pipeline. HuGaDB windows can use six IMU streams, whereas USC-HAD windows provide one IMU and are zero-padded to match the 30-D input dimension. That sensing-layout mismatch and label-definition mismatch are enough to produce visibly different accuracies under the same classifier family.

Another important point is that the binary task itself is intentionally simplified. The positive class is "active movement" rather than a specific gait mode, and HuGaDB window labels are assigned by majority vote over sample-level annotations. This design is appropriate for an assist on/off controller, but it limits how strongly the reported accuracy can be interpreted as full intention understanding. The present results therefore support robust *gating* of assistance, not full activity forecasting or terrain-aware torque generation.

For that reason, the final pipeline now prioritizes **methodological consistency** over the highest headline number. HuGaDB is used for training and pseudo-real-time replay without introducing a separate deploy-pool binary checkpoint. This avoids mixing public datasets with different sensing layouts in the deployed path and avoids relying on a separate phone-data prototype.

4.4 Multiclass Activity Recognition

Beyond the binary controller, the repository includes **dataset-native multiclass SVM** baselines (implemented with ECOC, no cross-dataset label unification): USC-HAD uses twelve trial-level activities; HuGaDB uses twelve per-sample activity IDs. Five-fold OOF accuracies on stratified subsamples are **53.41%** (USC-HAD) and **75.78%** (HuGaDB). Figure 4.2 shows the HuGaDB confusion matrix; both matrices are written by `EvaluateMulticlassConfusion` per dataset.

From a system-design perspective, the multiclass result should be interpreted as an exploratory baseline rather than as a deployable recognition layer. It demonstrates that the project has progressed beyond a simple binary detector and now has per-dataset native labels and a reproducible multiclass training path. However, the current accuracy is not yet sufficient for fine-grained activity-dependent torque scheduling. Additional class balancing and subject-wise validation would be required before multiclass predictions could be trusted for direct control decisions.

4.5 Real-Time Pipeline Behavior

Figure 4.3 presents the output of the online MATLAB pipeline on a held-out HuGaDB session, combining SVM classification, finite-state-machine (FSM) smoothing, and an optional Kalman-based kinematic visualization trace. The figure shows that the pipeline

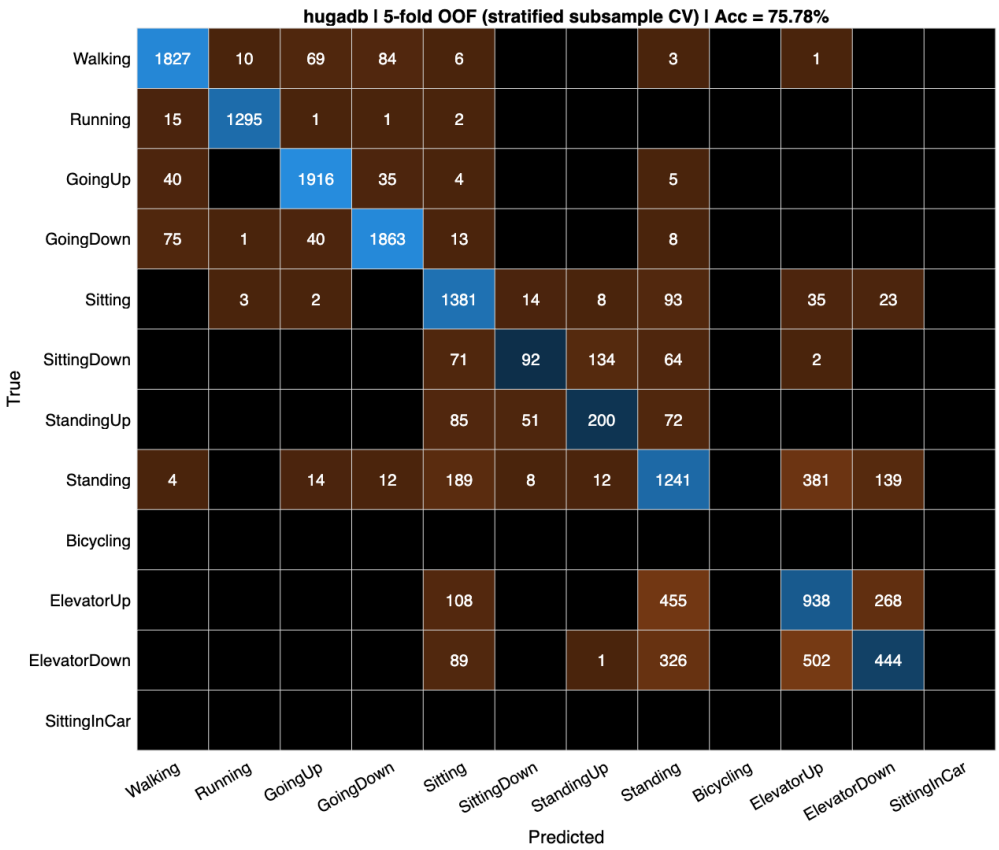


Figure 4.2: HuGaDB native twelve-class multiclass SVM confusion matrix (five-fold OOF, stratified subsample for tractable runtime).

produces a stable latched assist command rather than rapidly toggling in response to individual window predictions.

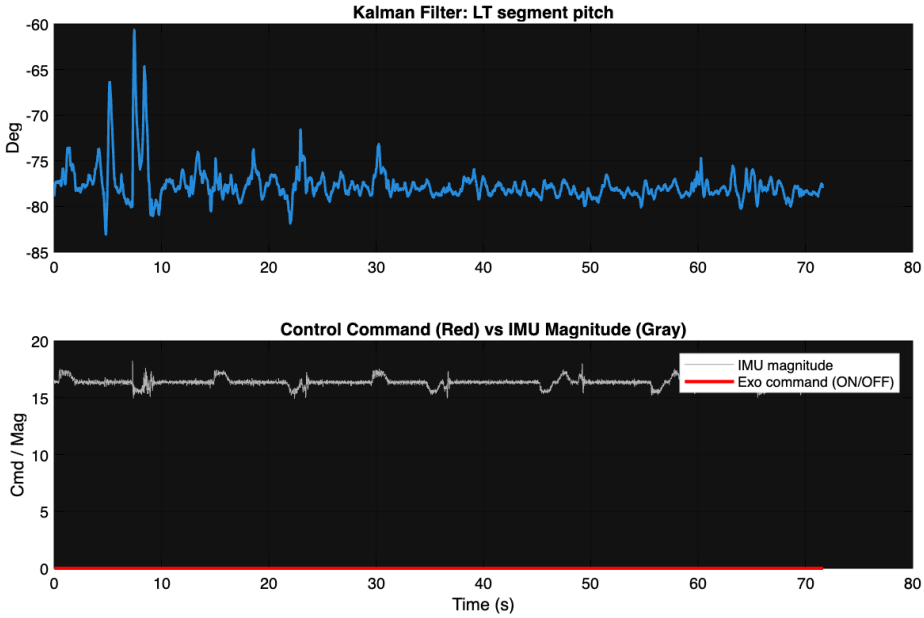


Figure 4.3: Real-time pipeline output on held-out HuGaDB replay: Kalman-smoothed segment pitch (top) and control command versus IMU magnitude (bottom).

This behavior is significant because the FSM design directly addresses a practical control issue: a high-accuracy classifier can still generate undesirable actuator chatter if predictions fluctuate near transition boundaries. Requiring three consecutive active predictions to enter assist and five consecutive inactive predictions to exit assist makes the controller conservative in a useful way. The trade-off is extra response delay, so the original 100–200 ms predictive-horizon claim should still be regarded as a design aspiration rather than a re-validated result of the submitted repository.

4.6 Sequence-Model Status

The repository now includes per-dataset LSTM baselines for the same pivot: binary sequence models on USC-HAD and HuGaDB, plus dataset-native multiclass sequence models on USC-HAD and HuGaDB. These are intentionally not treated as the primary reported result in this chapter because their current evaluation uses a stratified 20% holdout rather than the five-fold OOF protocol used for the SVM baselines.

Binary LSTM holdout accuracy is **99.68%** on USC-HAD and **98.36%** on HuGaDB. These numbers come from the saved per-dataset outputs `results/metrics/binary/lstm_evaluation_metric` and `results/metrics/binary/lstm_evaluation_metrics_hugadb.mat`.

Native multiclass LSTM holdout accuracy is **91.10%** on USC-HAD and **83.80%** on HuGaDB. Because both datasets still use separate native label spaces, these values should

be read as per-dataset summary metrics rather than as a class-by-class cross-dataset comparison.

The holdout confusion matrix for the optional LSTM baseline is shown in Figure 4.4.

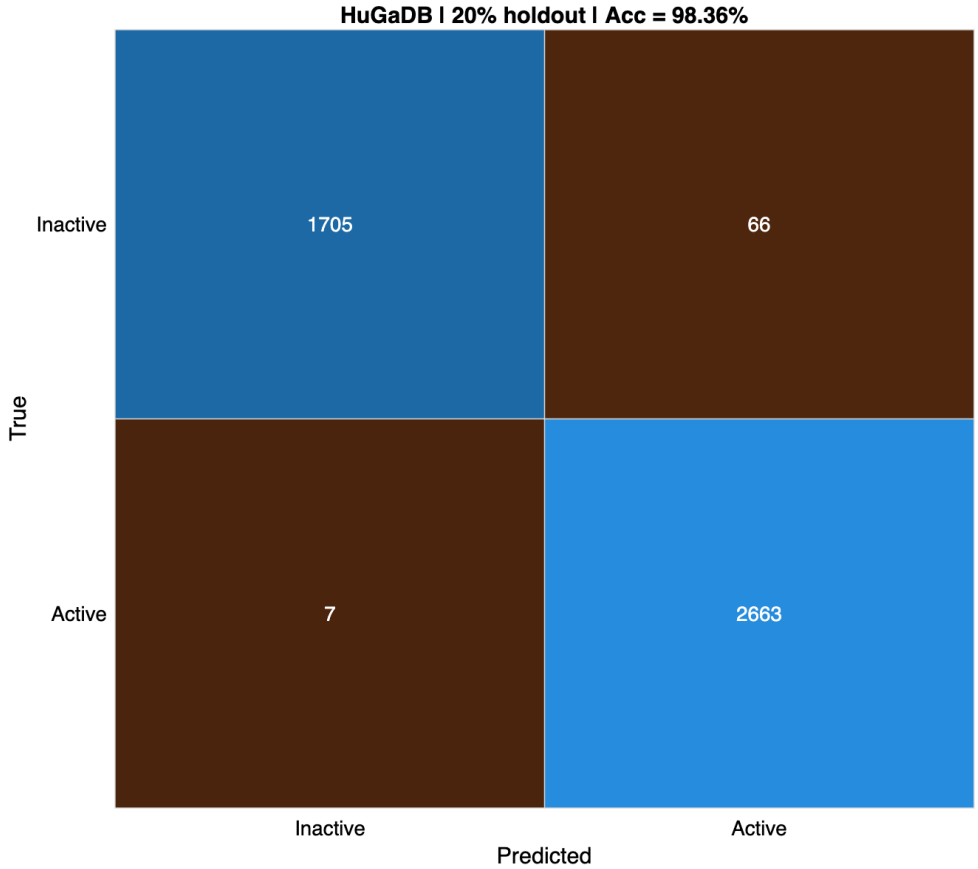
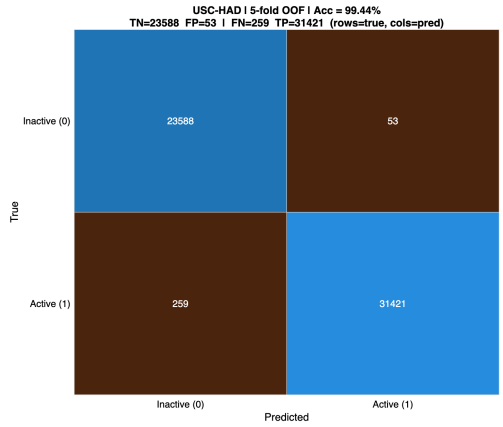


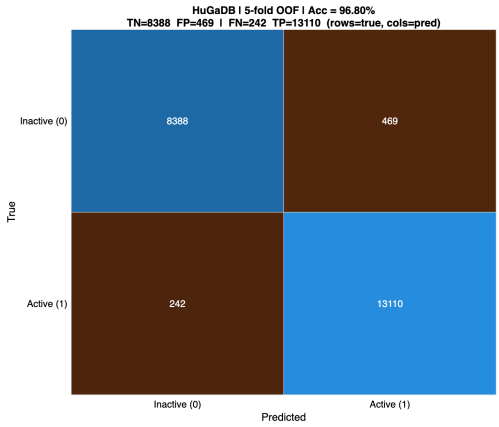
Figure 4.4: Optional HuGaDB binary LSTM confusion matrix on the stratified 20% holdout. When the network is fully trained and saved, predictions come from `EvaluateLstmConfusion`; if training was interrupted, the figure may be reconstructed from logged accuracy as described in the text.

4.7 Complete Generated Figure Gallery

For traceability, the report also includes the full set of distinct generated PNG artifacts currently written under `results/figures/`.

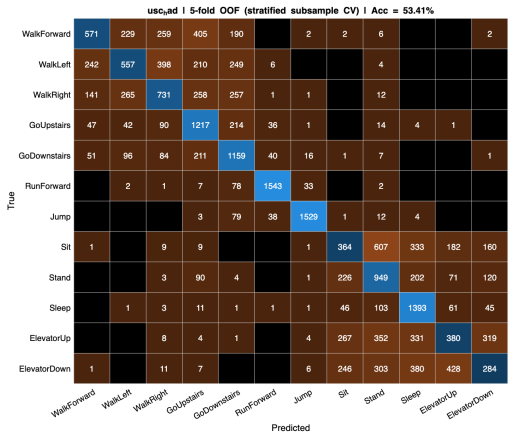


Binary SVM confusion matrix, USC-HAD.

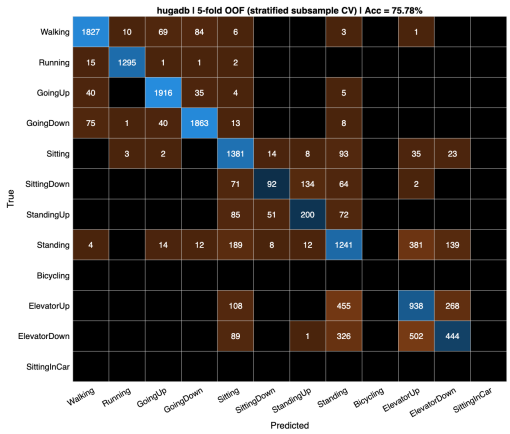


Binary SVM confusion matrix, HuGaDB.

Figure 4.5: Binary SVM benchmark comparison figures generated by `RunSvmDatasetAblation`.

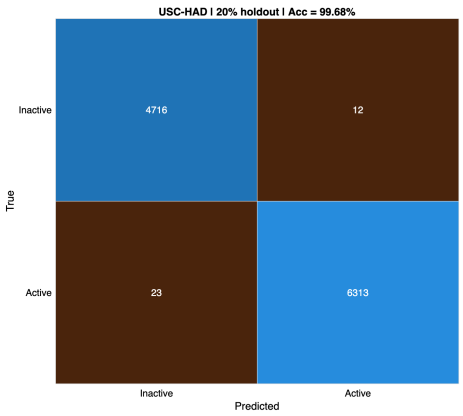


Native multiclass SVM confusion matrix, USC-HAD.

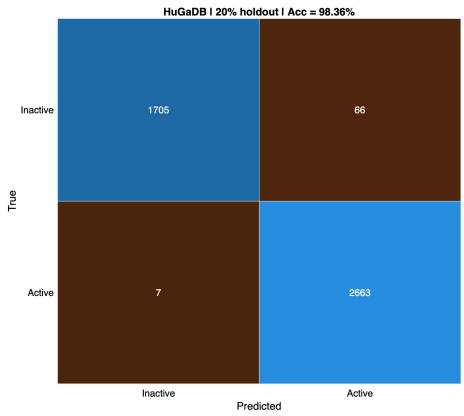


Native multiclass SVM confusion matrix, HuGaDB.

Figure 4.6: Dataset-native multiclass SVM figures generated by `RunTrainEvalMulticlass`.

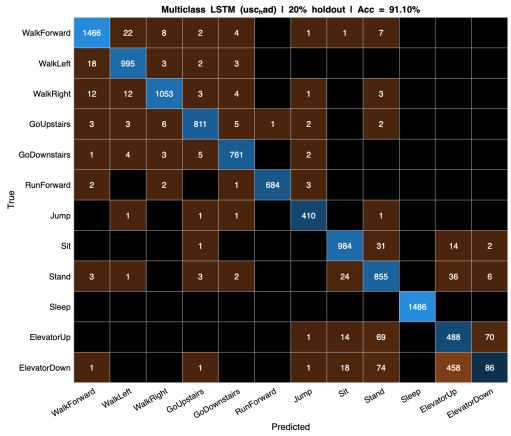


Binary LSTM confusion matrix, USC-HAD holdout split.

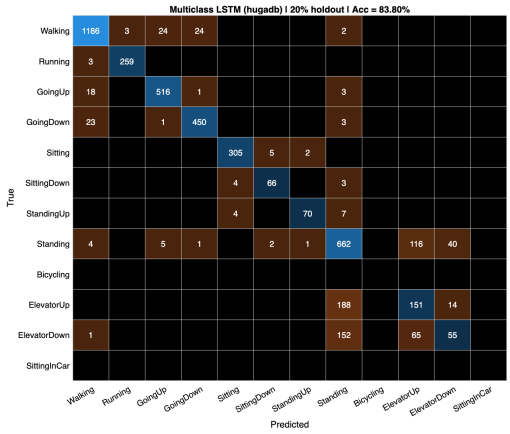


Binary LSTM confusion matrix, HuGaDB holdout split.

Figure 4.7: Binary LSTM figures generated by `RunLstmDatasetAblation`.

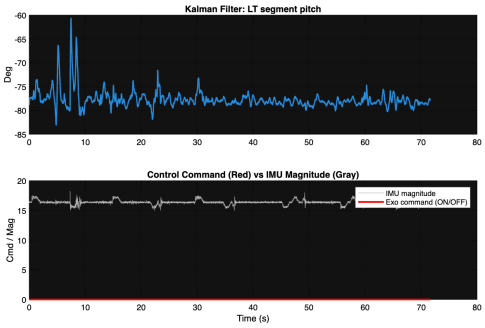


Native multiclass LSTM confusion matrix, USC-HAD holdout split.

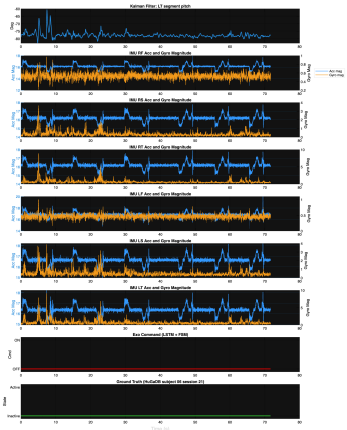


Native multiclass LSTM confusion matrix, HuGaDB holdout split.

Figure 4.8: Dataset-native multiclass LSTM figures generated by `RunTrainEvalLstmMulticlass`.

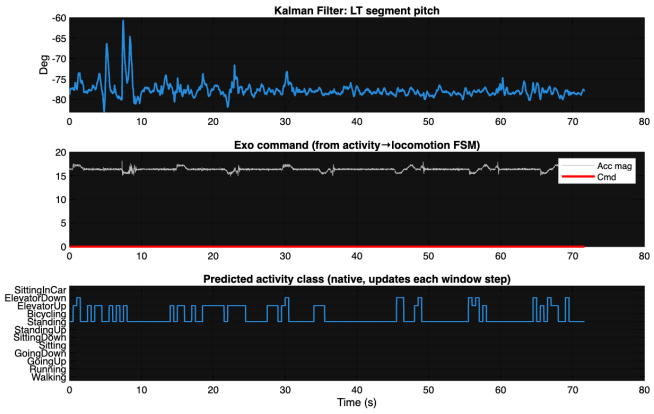


Binary SVM replay pipeline on held-out HuGaDB data.



Binary LSTM replay pipeline on held-out HuGaDB data.

Figure 4.9: Replay-pipeline figures generated by the binary SVM and binary LSTM scripts.



Multiclass SVM replay pipeline on held-out HuGaDB data.

Figure 4.10: Additional replay-pipeline figure generated by the multiclass SVM script.

4.8 Overall Interpretation

Taken together, the results show that the project has successfully delivered a reproducible software baseline for exoskeleton assist gating. The final repository now separates benchmark comparison from the actual deployment path: USC-HAD and HuGaDB are evaluated separately for comparison evidence, while the active binary SVM pipeline is based on HuGaDB and reserves subjects 06 and 08 for simulation. The Kalman and FSM blocks remain integrated into an end-to-end control flow, and the report metrics are traceable to versioned MATLAB outputs. At the same time, the discussion also makes clear what remains unfinished: subject-specific tuning, truly predictive transition detection, hardware deployment, and controlled user validation under realistic load-transport tasks.

4.9 Visualization Tools and Software Architecture

Real-time visualization is integrated in MATLAB through scripts such as `RunExoskeletonPipeline.m`.

The codebase uses per-script path resolution, a modular `src/` tree for acquisition, fusion, classification, and features, and `docs/System_Design.md` for architecture documentation. Repository: `https://github.com/Rex-Yim/AutomationForExoskeleton`.

Chapter 5

Conclusions and Future Work

5.1 Summary of what was delivered

This project produced a reproducible MATLAB baseline for lower-limb exoskeleton *assist gating* on public IMU data: a HuGaDB-centered replay path, optional Kalman visualization (`imufilter`), a 30-dimensional feature pipeline, binary RBF SVM classification with five-fold out-of-fold evaluation, optional native ECOC models (USC-HAD twelve classes; HuGaDB eight classes) for exploratory activity recognition, per-dataset binary LSTM sequence classifiers, per-dataset multiclass LSTM sequence baselines, and finite-state-machine smoothing with asymmetric hysteresis for stable on/off assist commands in simulation. Scalar accuracies and figures are traceable to versioned scripts and `results/metrics/*/*.mat`, with numbers exported into `generated_metrics.tex` via `ExportMetricsForReport.m`.

The benchmark comparison table still shows that the binary accuracy target ($> 90\%$) is met for USC-HAD training, while HuGaDB accuracy is lower, consistent with dataset heterogeneity and label mapping choices. The final repository trains on sixteen HuGaDB subjects and reserves subjects 06 and 08 for simulation. Multiclass accuracy remains modest and is not proposed as a deployable control layer without further work. End-to-end hardware trials, embedded deployment, hybrid SVM–LSTM fusion as in the original pseudocode, and the full predictive-horizon benchmark were not completed in software; the sections below preserve interim future-work bullets, timeline, risks, and repository status so planned scope stays distinguishable from the validated repository.

5.2 Future work beyond the FYP (interim PDF)

- **Personalization:** Short calibration walks with transfer learning and lightweight RL (e.g., PPO/SAC) to tune assistance.
- **Injury-risk assessment:** Integrate ergonomic risk frameworks such as Zelik et

al. [44] into user trials.

5.3 Project timeline (interim PDF)

Table 5.1 reproduces the phased plan from the Jan 2026 report (notional dates; confirm with your advisor).

Table 5.1: Phased development plan (interim submission).

Phase	**Duration**	**Key activities**	**Milestones**
ML training & simulation integration	Jan – Feb 2026 (2 months)	Train/deploy ML; Simulink HIL-style virtual tests; locomotion modes under load	Validated models ($\geq$ 90% target); simulation demo
Hardware integration & testing	Mar – Apr 15, 2026 (1.5 months)	Software–hardware integration; bench and limited treadmill tests (5–10 trials); ethics	Logs with latency/error metrics
Optimization, documentation, finalization	Apr 16 – May 2026 (1.5 months)	Refine from tests; compile report; presentation	Submitted report; presentation; archived repository

5.4 Risk assessment (interim PDF)

Table 5.2 summarizes risks and mitigations from the Jan 2026 report.

Table 5.2: Risks and mitigations (interim submission).

Risk	**Like.**	**Imp.**	**Mitigation**
Hardware integration delays (e.g., actuator vs. ExoTechHK frame)	M	H	Modular COTS parts; parallel software; early prototype access
ML underperformance (< 85% accuracy)	L	M	Rule/ML hybrid fallback; partner datasets; cross-validation / RL tuning
Trial delays	M	H	Public/simulated data interim; limit volunteers; secure IRB early
Budget overrun	L	L	University funding; HKSTP supplements

5.5 Repository status (consolidated, 2026)

Relative to the interim plan, the following are **delivered in software**: HuGaDB-based binary RBF SVM training; dataset ablation scripts that preserve USC-HAD /

HuGaDB comparison runs; optional Kalman visualization baseline; optional native ECOC SVMs per public dataset; binary and multiclass LSTM sequence classifiers per dataset (`TrainLstmBinary`, `EvaluateLstmConfusion`, `TrainLstmMulticlass`, `EvaluateLstmMulticlassConfu` `RunExoskeletonPipelineLstm`); asymmetric FSM; documentation and evaluation figures.

Not delivered: hybrid SVM+LSTM fusion as in the original pseudocode, MATLAB Coder deployment, ROS/PID torque layer, hardware controller integration, and prospective user trials—these remain aligned with the roadmap and risk tables above.

Synthesis: The LaTeX report intentionally preserves the **full interim narrative** (objectives, proposed LSTM/EKF/ROS extensions, timeline, risks) while adding a **faithful description of the implemented repository**, so readers can distinguish planned scope from validated baseline.

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